ANALYSTLAB AFRICA / HEALTHCONNECT

HealthConnect Appointment Attendance Analytics

Week 5 Analytics Report

Prepared by	Chenai Mpofu
Track	Data Analytics
Reporting period	January 2025 - June 2026
Document	Initial analytics report and Power BI dashboard narrative

KEY MESSAGE

Appointment non-attendance is the central operational risk: 51.2% of completed appointments were no-shows, with the strongest observed differences linked to previous no-show history, booking lead time and reminder status.

1. Executive summary

This Week 5 analysis extends the Week 4 data-quality assessment into a decision-focused review of appointment attendance. The cleaned dataset contains 5,000 appointments for 1,696 patients between January 2025 and June 2026. Power BI was used to create a date model, calculate attendance KPIs and examine patterns across time, booking lead time, reminder status, appointment type, appointment time and previous no-show history.

> Of the 4,737 attended or no-show appointments, 2,423 were no-shows. This produces a no-show rate of 51.2%. The 263 cancellations are tracked separately and represent 5.3% of all appointments.

Management interpretation

Primary risk indicators

- Previous no-show history is the clearest behavioural indicator in the descriptive analysis, with a no-show rate of 57.8% compared with 46.3% for patients without prior no-shows, an 11.5 percentage-point difference.
- Longer booking lead times are also associated with non-attendance. No-show appointments were booked 34.5 days ahead on average, about 10 days longer than attended appointments at 24.5 days.

Operational factors to monitor

- Appointments without a recorded reminder have a 54.6% no-show rate, compared with 49.9% where a reminder was sent.
- Follow-up appointments have the highest appointment-type no-show rate at 54.2%, while General Consultations have the lowest at 49.1%.
- Monthly no-show rates range from 44.1% to 57.6%, so performance should be monitored over time rather than judged from a single month.

Operational relevance

Capacity impact

The high no-show rate points to substantial underuse of available appointment capacity. Missed appointments disrupt clinic schedules and increase administrative workload.

Patient access impact

When booked slots go unused, waiting times can increase and other patients may have fewer opportunities to access care.

Management action

By tracking no-show patterns, HealthConnect can target reminders, improve appointment planning and use clinical capacity more effectively.

2. Transition from Week 4

Week 4 established that the file was structurally suitable for analysis, documented the fields and recorded the main data-quality risks. Week 5 builds on that foundation by formalising KPI definitions, creating a date dimension, testing patterns visually and translating findings into operational recommendations.

Week 4 foundation	Week 5 extension
Profiled 5,000 rows and 18 fields	Built reusable DAX measures and two report pages
Confirmed unique appointment IDs and no duplicate rows	Used appointment-level counts without adjustments
Documented missing distance and waiting-time values	Kept denominators explicit and avoided imputing unsupported values
Identified US-formatted source dates	Parsed dates using English (United States) locale and linked a Date Table
Separated Attended, No-Show and Cancelled outcomes	Excluded cancellations from the no-show-rate denominator and reported cancellation separately

3. Data preparation and analytical model

3.1 Source and scope

Item	Analytical scope
Rows	5,000 appointment records
Patients	1,696 unique patient IDs
Period	1 January 2025 to 30 June 2026
Outcome counts	2,423 No-Show; 2,314 Attended; 263 Cancelled
Primary tool	Microsoft Power BI Desktop
Grain	One row per appointment

3.2 Preparation completed

1. Loaded the appointment file and promoted the first row to column headers.
2. Reviewed data types, missing values, category consistency, duplicates and logical relationships between fields.

3. Converted booking_date and appointment_date to Date using English (United States) locale because the CSV used M/D/YYYY formatting.
4. Confirmed appointment_date was 100% valid after conversion.
5. Created a Date Table spanning the minimum and maximum appointment dates and added Year, Month, Month Number, Quarter and Year-Month fields.
6. Created an active many-to-one relationship from Appointments[appointment_date] to Date Table[Date], using single-direction filtering.
7. Sorted Month by Month Number and used Year-Month for chronological reporting across two calendar years.

3.3 Data-quality observations

8. The reminder_channel field has 1,366 missing values (27.3%); these records correspond to reminder_sent = No and are therefore structurally expected.
9. distance_to_clinic_km is missing for 90 appointments (1.8%).
10. waiting_time_minutes is missing for 60 appointments (1.2%).
11. No duplicate appointment IDs, invalid appointment dates or category inconsistencies were identified.
12. The source dictionary describes ISO dates, while the CSV is formatted as M/D/YYYY. Locale-aware parsing was therefore required.

4. KPI definitions and results

All percentage results are displayed to one decimal place in the dashboard. The underlying calculations retain full precision.

KPI	Definition	Result	Business question
Total Appointments	Count of all appointment rows	5,000	How much appointment activity occurred?
No-Show Appointments	Outcome = No-Show	2,423	How many booked appointments were missed?
No-Show Rate	No-Shows / (No-Shows + Attended)	51.2%	What share of completed attendance opportunities became no-shows?
Cancellation Rate	Cancelled / Total Appointments	5.3%	What share of all bookings were cancelled?
Reminder Coverage Rate	Reminder Sent = Yes / Total Appointments	72.7%	How consistently were reminders recorded as sent?
Repeat No-Show Rate	No-Shows among eligible appointments with previous_no_shows >= 1	57.8%	How frequently do patients with prior no-shows miss again?
Average Booking Lead Days	Average appointment date minus booking date	29.6 days overall	How far in advance are appointments booked?

NB: Cancelled appointments are intentionally excluded from the no-show-rate denominator because a cancellation is a known booking outcome rather than an unattended appointment. Cancellations are measured separately against all appointments.

5. Exploratory analysis and dashboard

5.1 Overview

The overview page combines the six headline KPIs with a monthly trend and two operational comparisons. Year-Month is used on the time axis so January 2025 and January 2026 remain separate and chronologically ordered. The Year-Month and Appointment Type slicers support interactive filtering across the dashboard.

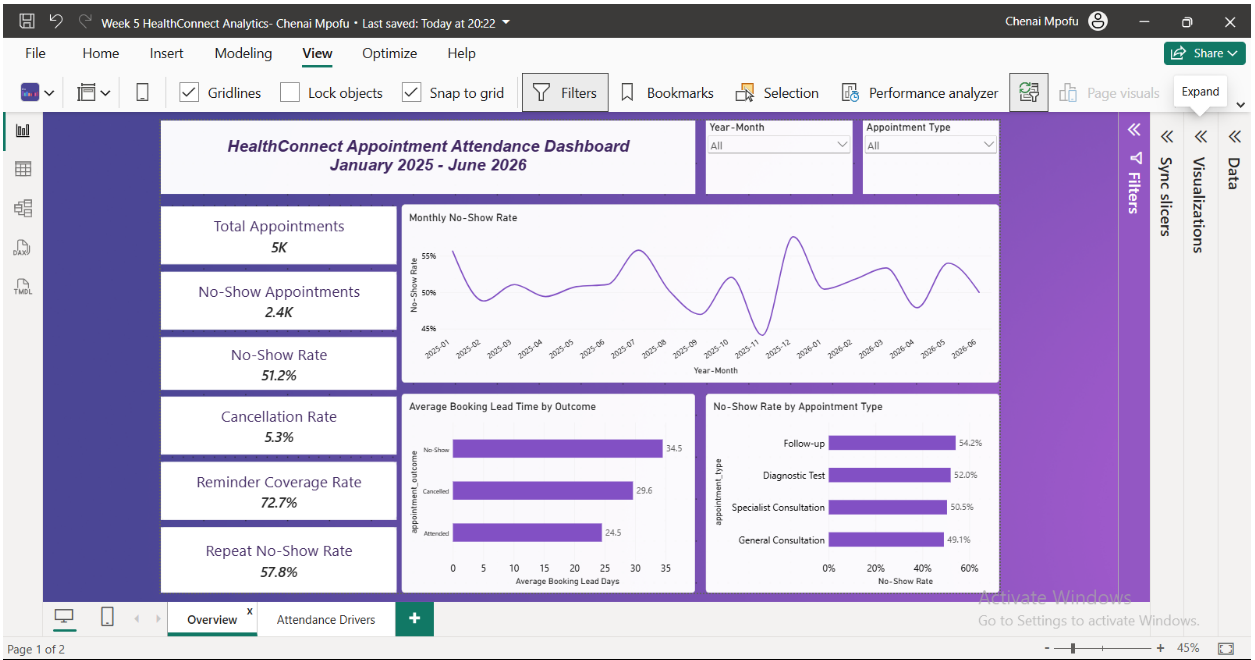

Figure 1. HealthConnect Appointment Attendance Dashboard - Overview page (Power BI).

5.2 Attendance Drivers

The second page focuses on descriptive factors associated with no-shows. It uses the same 51.2% overall no-show definition and compares groups within reminder status, appointment time and previous no-show history. The synchronized Year-Month and Appointment Type slicers allow the same segments to be reviewed consistently across both report pages.

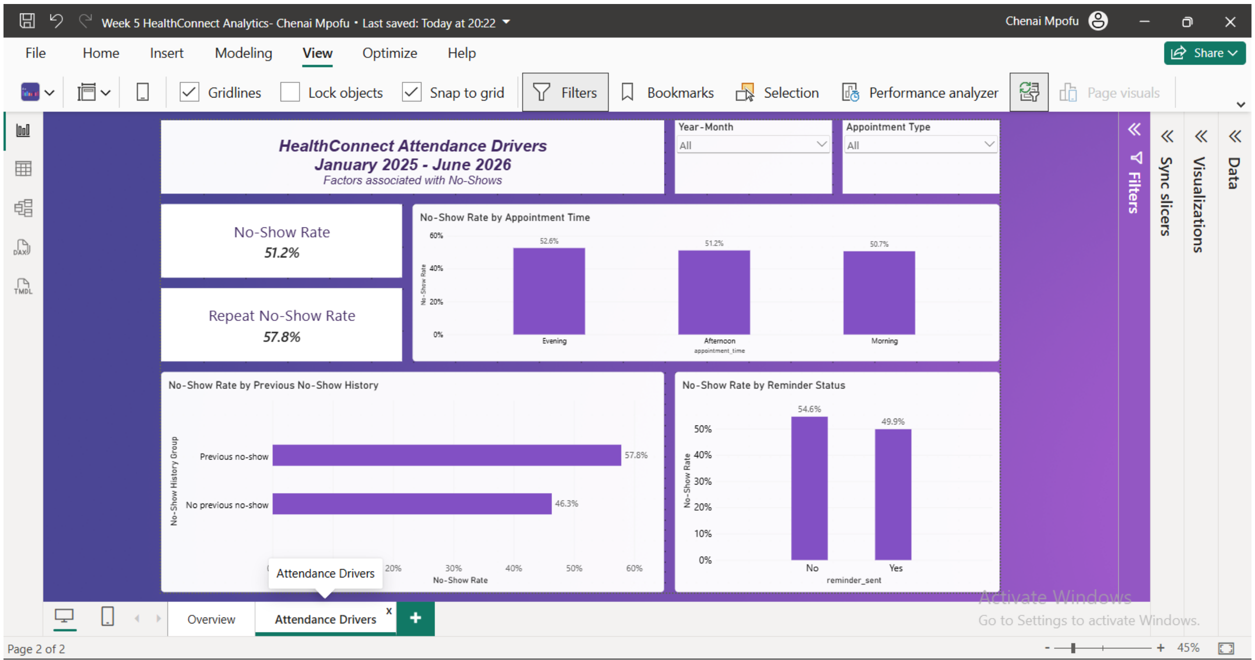

Figure 2. Attendance Drivers page - reminder status, appointment time and no-show history (Power BI).

6. Key insights and business implications

1. No-shows are the dominant attendance risk

Evidence: The no-show rate is 51.2%: 2,423 no-shows among 4,737 attended or no-show appointments.

Implication: More than half of attendance opportunities are being lost, creating pressure on capacity utilisation and service access.

2. Previous no-show history is the strongest observed separator

Evidence: Patients with at least one previous no-show have a 57.8% rate, compared with 46.3% for those without such history (+11.5 percentage points).

Implication: Prior behaviour can support risk-based follow-up and booking confirmation workflows.

3. Reminder status is associated with lower non-attendance

Evidence: The no-show rate is 54.6% without a recorded reminder and 49.9% when a reminder was sent (-4.8 percentage points).

Implication: Reminder coverage and delivery quality should be monitored, but the difference is observational and does not prove causation.

4. Longer booking lead time is linked to no-shows

Evidence: No-show appointments were booked 34.5 days ahead on average, compared with 24.5 days for attended appointments (+10.0 days).

Implication: Long-lead bookings may benefit from staged reconfirmation and easier rescheduling.

5. Appointment-type variation identifies a priority segment

Evidence: Follow-up appointments have the highest rate at 54.2%, followed by Diagnostic Tests at 52.0%, Specialist Consultations at 50.5% and General Consultations at 49.1%.

Implication: Follow-up workflows are a sensible first area for targeted testing.

6. Appointment time has a smaller but usable difference

Evidence: Evening appointments have a 52.6% rate, compared with 51.2% in the afternoon and 50.7% in the morning.

Implication: Time of day may refine risk prioritisation, although its observed effect is weaker than previous history.

7. Monthly rates are volatile

Evidence: Monthly rates range from 44.1% in November 2025 to 57.6% in December 2025.

Implication: Monthly monitoring is necessary to detect deterioration and evaluate whether interventions are sustained.

7. Recommendations

Priority action	Implementation rationale
Prioritise high-risk confirmations	Create a workflow for patients with previous no-shows, beginning with a confirmation request and an easy rescheduling option.
Increase and audit reminder coverage	Move beyond the current 72.7% coverage by tracking reminder delivery, timing, channel and response rather than only whether a reminder was recorded.
Use staged reminders for long-lead bookings	For appointments booked well in advance, add a reminder or reconfirmation closer to the appointment date.
Pilot with follow-up appointments	Start intervention testing in the appointment category with the highest observed no-show rate, then compare results with a suitable baseline.
Maintain a rapid-fill wait-list	Use cancellations and early rescheduling signals to offer released capacity to patients who can attend at short notice.
Build a measurement plan	Monitor no-show rate, cancellation rate, reminder coverage, rescheduling rate and recovered appointment capacity by month.

8. Assumptions, limitations, risks and dependencies

Area	Current position	Effect on interpretation
Observational analysis	The data describes associations; no randomised intervention was conducted.	Reminder and lead-time differences cannot be interpreted as causal effects.
Dataset scope	The analysis uses one supplied HealthConnect dataset and no external benchmark.	Results may not generalise to other clinics, regions or periods.
No-show definition	Cancellations are excluded from the no-show denominator.	The KPI is consistent and transparent, but it should not be compared with differently defined external rates.
Missing values	Distance is missing in 1.8% and waiting time in 1.2% of records.	Analyses using those fields require explicit handling and denominator disclosure.
Reminder detail	Reminder timing, delivery success, response and failed-contact reasons are unavailable.	The reminder comparison is limited to recorded sent/not sent status.
Waiting time	The field is described as estimated; the point at which it was estimated is not specified.	It should not be presented as a confirmed patient experience measure.
Repeated patients	Multiple appointments may belong to the same patient.	Appointment-level observations may not be statistically independent.

Area	Current position	Effect on interpretation
Operational context	Clinic capacity, staffing, costs, transport barriers and no-show reasons are unavailable.	Commercial impact cannot yet be converted into a precise financial value.

Dependencies for the next phase

13. Agreement with stakeholders on the operational definition of no-show and the acceptable target rate.
14. Access to reminder delivery timestamps and delivery-status data.
15. Operational ownership for confirmation, rescheduling and wait-list processes.
16. A controlled method for assessing whether an intervention changes outcomes.
17. Data Science review before any predictive model is developed or deployed.

9. Cross-track collaboration

A request for Data Science input has been posted to the shared project channel. The response is pending. No feedback or modelling decision is attributed to another intern in this draft.

The request asks whether previous no-show history, booking lead time and reminder status are appropriate candidate features for predictive modelling, and requests one recommendation or limitation to consider.

10. Conclusion and next steps

The analysis establishes a clear baseline: appointment non-attendance is high and is most strongly differentiated by prior no-show history, booking lead time and reminder status. The next phase should turn these descriptive findings into testable operational interventions, strengthen data capture and evaluate outcomes over time.

1. Pilot enhanced reminders and reconfirmation for patients with previous no shows, long booking lead times and follow-up appointments.
2. Monitor no-show rates monthly using consistent KPI definitions.
3. Collect reminder- delivery, rescheduling and capacity- recovery data to support stronger operational evaluations.
4. Consider predictive modelling only after appropriate governance, fairness, validation and workflow ownership are established.

Appendix A. Measure logic

Measure	DAX logic (abridged)
Total Appointments	COUNTROWS(Appointments)
No-Show Appointments	CALCULATE([Total Appointments], appointment_outcome = "No-Show")
Attended Appointments	CALCULATE([Total Appointments], appointment_outcome = "Attended")
No-Show Rate	DIVIDE([No-Show Appointments], [No-Show Appointments] + [Attended Appointments], 0)
Cancelled Appointments	CALCULATE([Total Appointments], appointment_outcome = "Cancelled")
Cancellation Rate	DIVIDE([Cancelled Appointments], [Total Appointments], 0)
Reminders Sent	CALCULATE([Total Appointments], reminder_sent = "Yes")
Reminder Coverage Rate	DIVIDE([Reminders Sent], [Total Appointments], 0)
Average Booking Lead Days	AVERAGE(Appointments[booking_lead_days])

Source: HealthConnect_Appointment_Data.csv and HealthConnect_Data_Dictionary.xlsx. Analysis prepared in Microsoft Power BI Desktop.